CPEN 416 / EEСE 571O

Lecture 17: manipulating density matrices; quantum channels

Thursday 6 November 2025

Announcements

- No office hours tomorrow (travelling)
- Reading break next week - no tutorial, no class.
  - Next class Thurs. 13 Nov.
  - Next tutorial Mon. 17 Nov.
  - Next quiz Tues. 18 Nov. (L16-L18)
- Bonus reading break assignment on Piazza
- Remember to fill out weekly project surveys

Last time

We introduced observables

$M = M^\dagger$ Hermitian

We measure an observable by associating its eigenvectors with a basis and assigning real values (its eigenvalues) as outcomes.

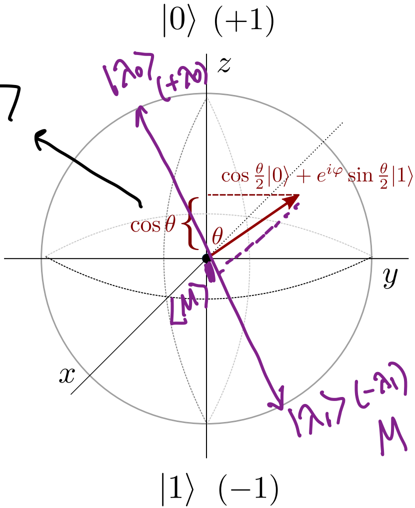
$Z: \; |0\rangle \to +1$
$\;\;\;\;\; |1\rangle \to -1$
eigenvectors. eigenvals.

The *expectation value* of an observable is

$\langle M \rangle = \langle \psi | M | \psi \rangle$

Last time

We associated expectation values of Paulis to the Bloch vector.



$\langle Z\rangle$

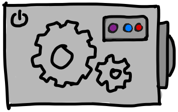
qml. expval

Last time

We introduced *mixed states*.

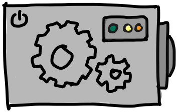
$$|0\rangle\langle 0| = \begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}$$

H basis: always $|+\rangle$



.... $|+\rangle$ .... $|+\rangle$ .. $|+\rangle$ ... $|+\rangle$ = one ket

pure state



.... $|0\rangle$ .... $|1\rangle$ .. $|1\rangle$ ... $|0\rangle$

$\frac{1}{2}$ $|+\rangle$

$\frac{1}{2}$ $|-\rangle$

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$$

Mixed states are probabilistic mixtures of pure states,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \qquad \sum_i p_i = 1$$

Learning outcomes

- Apply unitary operations to mixed states
- Perform measurements on mixed states
- Define and apply quantum channels to quantum states

Density matrices

Density matrices of mixed states are linear combinations of those of pure states:

$$\rho = \sum_i p_i \underbrace{|\psi_i\rangle\langle\psi_i|}_{\rho_i \text{ for pure state } |\psi_i\rangle} \qquad \sum_i p_i = 1$$

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

$$\rho = \frac{1}{3}\rho_+ + \frac{2}{3}\rho_0 = \frac{1}{6}\begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{3}|+\rangle\langle+| + \frac{2}{3}|0\rangle\langle 0|$$

$$= \frac{1}{3}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$= \frac{1}{6}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{2}{3}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\rho_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\rho_- = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Density matrices

$A = \sum_i \lambda_i |\lambda_i\rangle\langle\lambda_i|$
(spectral theorem)

Density matrices have some nice properties.

- they are Hermitian $\rho = \rho^\dagger$
- they have trace 1 → sum of diagonal
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

Check with our example:

$$\rho = \frac{1}{6}\begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}$$

$\lambda_0 = 0.87\ldots$
$\lambda_1 = 0.127\ldots$

Fun activity: show properties hold for general $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$

Unitary operations and mixed states

initial $\rho = \frac{1}{6}\begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{3}|+\rangle\langle+| + \frac{2}{3}|0\rangle\langle 0|$

Exercise: if H to our mixed state ($|+\rangle$ w/prob. 1/3, $|0\rangle$ w/prob. 2/3), what is the new state?

Apply H: $\rho' = \frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|+\rangle\langle +|$

$= \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$

$\rho \xrightarrow{???} \rho'$

$H\rho = \frac{1}{3}|0\rangle\langle +| + \frac{2}{3}|+\rangle\langle 0|$

$H\rho H^\dagger = \frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|+\rangle\langle +|$

Unitary operations and mixed states

For pure state $|\psi\rangle$ and operation U,

$$|\psi\rangle \rightarrow |\psi'\rangle = U|\psi\rangle$$

As mixed states,

$$|\psi\rangle\langle\psi| \longrightarrow |\psi'\rangle\langle\psi'| = \left(U|\psi\rangle\right)\left(\langle\psi|U^\dagger\right)$$
$$= U|\psi\rangle\langle\psi|U^\dagger$$

More generally,

$$\rho = \sum p_i |\psi_i\rangle\langle\psi_i| \qquad \sum_i p_i = 1$$

$$\Rightarrow \rho' = \sum p_i \, U|\psi_i\rangle\langle\psi_i|U^\dagger$$
$$= U\left(\sum p_i |\psi_i\rangle\langle\psi_i|\right)U^\dagger$$
$$= U\rho U^\dagger$$

Mixed states and measurements

Recall that for a pure state $|\psi\rangle$, the probability of measuring and observing it in state $|\varphi\rangle$ is

$$\Pr(\varphi) = |\langle\varphi|\psi\rangle|^2$$

We can rewrite this...

$$\begin{aligned} |\langle\varphi|\psi\rangle|^2 &= \langle\varphi|\psi\rangle\langle\psi|\varphi\rangle \\ &= \langle\psi|\varphi\rangle\langle\varphi|\psi\rangle \\ &= \langle\psi| \underbrace{\left(|\varphi\rangle\langle\varphi|\right)}_{\rho_\varphi} |\psi\rangle \end{aligned}$$

$$\rho_\varphi^2 = \rho_\varphi$$

$\Pi_\varphi = |\varphi\rangle\langle\varphi|$ is the density matrix of $|\varphi\rangle$, which is a *projector*.

Mixed states and measurements

In a projective measurement, the $\{\Pi_k\}$ "project" a state down to the consituent eigenstate.

Example: Apply projector Π_0 to $|+\rangle$:

$$\Pi_0 = |0\rangle\langle 0|$$

$$\Rightarrow \Pi_0|+\rangle = \underline{\underline{|0\rangle}}\underbrace{\langle 0|+\rangle}_{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}|0\rangle$$

Apply projector again:

$$\Pi_0\Pi_0|+\rangle = |0\rangle\langle 0|\left(\frac{1}{\sqrt{2}}|0\rangle\right) = \frac{1}{\sqrt{2}}|0\rangle$$

probability: $\langle +|\Pi_0|+\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\end{pmatrix} = \frac{1}{2}$

Mixed states and measurements

proj. meas: $|\langle \varphi_i | \psi \rangle|^2 = \langle \psi | \Pi_i | \psi \rangle$

For mixed states, measurement follows the **Born rule**:

$$\Pr(\text{outcome } i) = \mathrm{Tr}(\Pi_i \rho)$$

where $\{\Pi_i\}$ is a **positive operator-valued measure (POVM)**.

The POVM elements satisfy

$$\sum_i \Pi_i = I$$

You can show this is equivalent to a projective measurement when ρ is pure.

Mixed states and measurements

$Tr(A) + Tr(B) = Tr(A+B)$

Example: $\{|+\rangle\langle+|, |-\rangle\langle-|\}$.

Given a state ρ,

$$Pr(+) = Tr(|+\rangle\langle+| \cdot \rho)$$

$$Pr(-) = Tr(|-\rangle\langle-| \cdot \rho)$$

$$|+\rangle\langle+| + |-\rangle\langle-| = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$$

Exercise: Show that $\{|+\rangle\langle+|, |-\rangle\langle-|\}$ form a legit POVM.

$$Tr(|+\rangle\langle+| \rho) + Tr(|-\rangle\langle-| \rho)$$

$$= Tr(|+\rangle\langle+| \rho + |-\rangle\langle-| \rho)$$

$$= Tr((|+\rangle\langle+| + |-\rangle\langle-|) \rho)$$

$$= Tr(I \cdot \rho)$$

$$= Tr(\rho)$$

$$= 1$$

Mixed states and measurements

Exercise: Suppose we prepare our system in $|+\rangle$ with probability $1/3$ and $|0\rangle$ with probability $2/3$. What is the probability of obtaining the POVM outcome $\Pi_+ = |+\rangle\langle+|$?

Mixed states and the Bloch sphere

What about expectation values? For pure states $|\psi\rangle$,

$$\langle X \rangle = \langle \psi | X | \psi \rangle$$

actual Pauli X

For a mixed state,

$$\langle X \rangle = 1 \cdot \Pr(+) + (-1) \cdot \Pr(-)$$

$$= 1 \cdot \mathrm{Tr}(|+\rangle\langle +| \rho) + (-1) \cdot \mathrm{Tr}(|-\rangle\langle -| \rho)$$

$$= \mathrm{Tr}\left[(|+\rangle\langle +| - |-\rangle\langle -|) \rho \right]$$

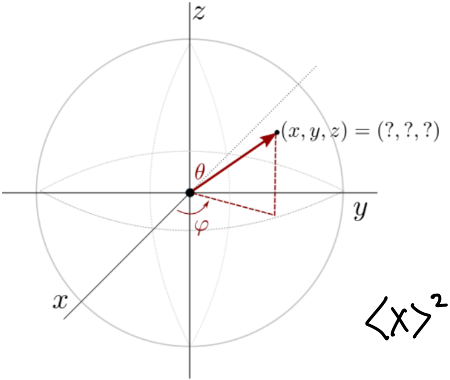
$$\sum_i \lambda_i |\lambda_i\rangle\langle \lambda_i|$$

$$= \mathrm{Tr}\left[\underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{=X} \rho \right] = \mathrm{Tr}(X\rho)$$

Mixed states and the Bloch sphere

Practice problem set 4...

Write a function that uses quantum measurements to extract its Bloch vector, i.e., its coordinates in 3-dimensional space, as demonstrated by the following graphic:



$\langle X \rangle = \mathrm{Tr}(X\rho)$

$\langle Y \rangle = \mathrm{Tr}(Y\rho)$

$\langle Z \rangle = \mathrm{Tr}(Z\rho)$

$\langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2 \leq 1$

Mixed states and the Bloch sphere

Given $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$, can we determine ρ? ☆ review on your own (didn't cover in class)

Recall ρ is Hermitian; Paulis are a basis.

$$\rho = a_I I + a_X X + a_Y Y + a_Z Z$$

ρ must have trace 1:

$$\mathrm{Tr}(\rho) = \mathrm{Tr}(a_I \underbrace{I}_{\mathrm{Tr}=2} + a_X \underbrace{X}_{\mathrm{Tr}=0} + a_Y \underbrace{Y}_{\mathrm{Tr}=0} + a_Z \underbrace{Z}_{\mathrm{Tr}=0})$$

$$1 = a_I \cdot 2$$

$$\Rightarrow a_I = \frac{1}{2}$$

Trace out another Pauli:

$$\mathrm{Tr}(X\rho) = \mathrm{Tr}(\cancel{a_I X} + a_X I + \cancel{a_Y (XY)} + \cancel{a_Z (XZ)})$$

$$= a_X \cdot 2 \quad \Rightarrow a_X = \frac{\mathrm{Tr}(X\rho)}{2} = \frac{\langle X \rangle}{2}$$

Mixed states and the Bloch sphere

More formally, we can write any ρ as

$$\rho = \frac{1}{2} I + a_X X + a_Y Y + a_Z Z$$

where $a_P = \mathrm{Tr}(P\rho) = \langle P \rangle$.

(Note that all of this generalizes to multiple qubits as well)

Mixed states and the Bloch sphere

Re-express:

$$\rho = \frac{1}{2}I + \frac{\langle X\rangle}{2}X + \frac{\langle Y\rangle}{2}Y + \frac{\langle Z\rangle}{2}Z$$

$$= \frac{1}{2}\begin{pmatrix} 1+\langle Z\rangle & \langle X\rangle - i\langle Y\rangle \\ \langle X\rangle + i\langle Y\rangle & 1-\langle Z\rangle \end{pmatrix}$$

Exercise: As ρ is positive semidefinite, its eigvals are ≥ 0. What constraint does this put on $\langle X\rangle, \langle Y\rangle, \langle Z\rangle$? (Hint: look at $\det(\rho)$)

$$\det(\rho) = \frac{1}{2}\left[(1+\langle Z\rangle)(1-\langle Z\rangle) - (\langle X\rangle - i\langle Y\rangle)(\langle X\rangle + i\langle Y\rangle)\right]$$

$$= \frac{1}{2}\left[1 - \langle Z\rangle^2 - \langle X\rangle^2 - \langle Y\rangle^2\right]$$

$$\geq 0 \quad \Rightarrow \quad \langle X\rangle^2 + \langle Y\rangle^2 + \langle Z\rangle^2 \leq 1$$

Mixed states and the Bloch sphere

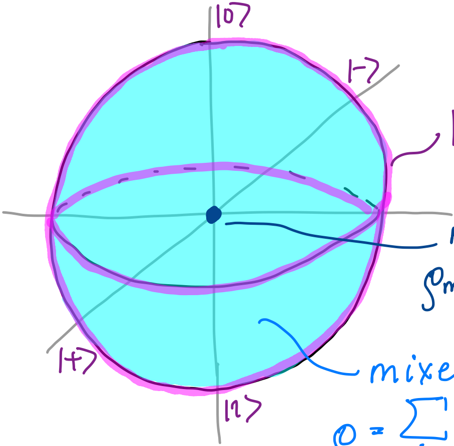
Exercise: How many eigenvalues does ρ have if it is pure? What constraint does this put on $\langle X\rangle, \langle Y\rangle, \langle Z\rangle$?

ρ pure $\longrightarrow$ $\rho = |\psi\rangle\langle\psi|$ $\rightarrow$ 1 eig. val. is non-zero

$\Rightarrow \det(\rho) = 0$

$\langle X\rangle^2 + \langle Y\rangle^2 + \langle Z\rangle^2 = 1$ is satisfied (surface of sphere!)

Mixed states live *in* the Bloch sphere!



pure states

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

maximally mixed state

$\rho_{max} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\mathbb{1}$

mixed states

$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$

Next time

Content:

- Error channels
- Noise in quantum systems

Action items:

1. Work on project (fill out surveys!)
2. Bonus reading break assignment

Recommended reading:

- From this class: Codebook NT; N & C 2.2.6, 2.4
- For next class: Codebook module NT, D; N & C 8.2-8.3,